

Toward a Theory of Computation for Learned Transformer Machines: Tasks, Resources, Learnability, Composition, and Closure

Paper 0.1 — Position and Formal Framework

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Abstract

Classical expressivity asks what an idealized Transformer can compute; benchmark studies ask whether a trained model succeeds. Between them lies the theory of a *learned* Transformer machine. We formalize task structure, information sufficiency, machine resources, acquisition probability, one-pass and autoregressive composition, and resource-scheduled closure. The operational object is compound, $\mathcal{M} = (F_\theta, C, S, I, \mathbb{T}, \mathcal{B})$. Five executable fixed-weight witnesses calibrate representability under explicit finite encodings. Their contrast with Paper 0.85—where trained one-token and autoregressive machines, including corrected data interventions, share frontier $K = 3$ and have zero three-seed acquisition at unseen $K = 4$ —makes the central distinction concrete: representability is not reliable acquisition. Empirical papers appear only as late, nonessential anchors.

1 Introduction

A fixed architecture may represent a computation that training rarely discovers; a trained model may fit bounded instances without a reusable rule; a reusable local rule may still fail under composition or termination. Autoregression changes the effective machine by applying the same transition repeatedly. We therefore separate

representability $\neq$ learnability $\neq$ reliable acquisition $\neq$ systematic executability $\neq$ closure.

The evidential layers are distinct: propositions follow from premises; executable hand-weighted constructions establish finite-domain representability; trained-model experiments measure acquisition; conjectures remain prospective.

2 Task families and concrete examples

A task family is $\mathcal{T} = (\mathcal{X}, \mathcal{Y}, \mathcal{D}, \mathcal{R})$, where $\mathcal{D}$ is a generator/distribution and $\mathcal{R}$ the declared prior. Its partially ordered complexity vector is

$$\mathbf{c} = (s_{\text{atom}}, k_{\text{comp}}, b_{\text{branch}}, v_{\text{state}}, n_{\text{nuisance}}, m_{\text{context}}, \dots).$$

The coordinates measure atomic syntax, composition/proof depth, branching, state, nuisance, and context/memory demand. Raw length need not increase composition depth.

2.1 From concrete tasks to the abstract task vector

T1: atomic mapping. $A \rightarrow B$, $B \rightarrow C$, QUERY B targets C: very low syntax, $k = 1$, unit branching, negligible state/nuisance.

T2: contextual lookup. $A \rightarrow X$, $B \rightarrow Y$, $C \rightarrow Z$, QUERY B targets Y: low syntax and $k = 1$, but contextual cross-token transport. Distractors can increase length without depth.

T3: grandparent. PARENT C B, PARENT B A, QUERY GRANDPARENT C targets A: parent lookup composed twice.

T4: root. For $C \rightarrow B \rightarrow A \rightarrow R$, a one-token answer needs bounded internal composition; multi-token C B A R needs recurrence and termination for scalable execution.

T5: ancestor-(k). ANCESTOR C 2 needs node-and-counter state, decrement, and termination.

T6: proof chain. From A, A->B->C->D, decide D?; proof depth, distractors, and branching vary independently.

T7: unification. $P(f(x), y)$ with $P(f(a), b)$ yields $\{x = a, y = b\}$: high syntax, binding state, and functor depth.

T8: branching search. A->B, A->C, B->D, C->E, ... separates search width and frontier/visited state from proof depth.

Task	s_{atom}	k_{comp}	b_{branch}	v_{state}	Context	Recurrence
Successor	low	1	1	low	optional	no
Pair lookup	low	1	1	low	yes	no
Grandparent	low	2	1	low	yes	no
Root	low	variable	1	current node	yes	useful
Ancestor-(k)	low	variable	1	node+counter	yes	yes
Proof chain	medium	variable	variable	proof state	yes	useful
Unification	high	bounded	1	bindings	yes	maybe
Branching search	medium	variable	high	frontier+visited	yes	yes

Table 1: Concrete tasks mapped to the abstract vector; entries are task requirements or useful protocols, not architectural lower bounds.

3 Information sufficiency

For query X , context D , target Y , and prior $\mathcal{R}$, validate

$$I(Y; D | X) = H(Y | X) - H(Y | X, D), \quad H(Y | X, D, \mathcal{R}) = 0$$

for deterministic identifiable tasks. Record

$$\mathcal{I} = (H(Y | X), H(Y | X, D), I(Y; D | X), \mathcal{R}, A_{\text{Bayes}}, |\mathcal{H}_{X, D, \mathcal{R}}|, m_{\text{id}}).$$

These mean, respectively: query-only uncertainty; remaining uncertainty after context; context’s added information; declared hypothesis family; best possible expected accuracy; number of consistent hypotheses; and minimum context for identification.

Impossible random mapping. A->7, B->2, QUERY C is underdetermined if C’s output is independent: $H(Y | X, D) > 0$.

Permutation completion. A->3, B->1, C->4, QUERY D forces D->2 only under the prior that outputs permute $\{1, 2, 3, 4\}$. This is constraint completion, not general induction.

Affine induction. Under $f(x) = ax + b \pmod{7}$, $f(0) = 3, f(1) = 5$ give $b = 3, a = 2$. The prior changes identifiability.

Standard Bayes-risk fact. Positive conditional entropy on positive measure implies $A_{\text{Bayes}} = \mathbb{E} \max_y P(Y = y | X, D, \mathcal{R}) < 1$. This routine fact enforces dataset discipline; it is not a novelty claim.

The zero-conditional-entropy condition applies to deterministically identifiable tasks, which Papers 0.5–0.9 primarily study; it is not a universal requirement for useful prediction. In a probabilistically predictive task, $H(Y | X, D, \mathcal{R}) > 0$: context can reduce uncertainty, $I(Y; D | X) > 0$, without eliminating it. Evaluation must then declare a null predictor, measure improvement over it, and compare with the Bayes-optimal distribution. This is a secondary evaluation regime, not a seventh computational axis; Appendix B gives the details.

4 Machine, controller, and resource formalism

C_0 invokes F_θ once; C_1 invokes it autoregressively; C_2 gives state typed fields; C_3 is an external action/tool loop.

Definition 1 (Compound learned machine).

$$\mathcal{M} = (F_\theta, C, S, I, \mathbb{T}, \mathcal{B}), qquad P_{\text{acquire}} = P_\Pi[A(\tau, \mathbf{c}; \mathcal{M}, \Pi) \geq \gamma].$$

F_θ is the learned transition map, C its scheduling protocol, S writable state, $I : q \mapsto D$ information, $\mathbb{T}$ partial external operations, and $\mathcal{B}$ context/call/token/time/memory budget.

Report $\Delta\mathcal{M} = (\Delta C, \Delta S, \Delta I, \Delta\mathbb{T}, \Delta\mathcal{B}, \Delta\Pi)$, so easier prompts, stronger tools, more time, and different training are not all “augmentation.”

5 Six computational axes

1. **Atomic acquisition:** which local operators pass across seeds?
2. **One-pass composition:** $K_{\text{internal}} = \max\{K : A(f^K) \geq \gamma\}$, controlling dependency radius, depth, nuisance, and length.
3. **Autoregressive composition:** $s_{t+1} = F_\theta(X, s_{\leq t})$; report transitions, stopping, sequence success, and train-short/test-long behavior.
4. **Closure signature:** separate a contiguous tested frontier from a uniform resource-scheduled procedure.
5. **In-context information:** separate copying, completion, task selection, parameter estimation, and procedure induction.
6. **Tool/harness augmentation:** separate computational, acquisition, reliability, and efficiency gains.

Much classical circuit/logical work studies a fixed one-pass Transformer under precision assumptions. It cannot be transferred automatically to C_1 , where emitted tokens create serial applications of the transition map.

6 Constructive witnesses and the learnability gap

Representability asks whether $\exists W^*$ such that $\mathcal{M}(W^*)$ implements τ ; acquisition asks for $P_\Pi[\mathcal{M}(W_\Pi)$ passes $\tau]$. The hand-weighted witnesses instantiate the first statement without training.

Witness	Protocol	((L,H,d))	Legal result	Main control	Scope
Successor	C_0	(1, 0, 4)	3/3, margin 1	SA 0/3	local weight map
Pair lookup	C_0	(1, 1, 9)	18/18	FF 6/18 ties	atomic pairs
Grandparent	C_0	(2, 1, 20)	2/2	one-layer SA 0/2	fixed two-hop chain
Root	C_1	(1, 1, 15)	24/24	FF 6/24	depth 0–3
Unary chain	C_1	(1, 1, 15)	96/96	FF 24/96	length 1–4

Table 2: Fixed-weight calibration. Exact means exhaustive argmax correctness, not one-hot attention; no witness is claimed minimal.

Paper 0.85 supplies the learned contrast. On its corrected Stage-1 dataset, one-token O0, free-trace O1, and supervised structured O3 have contiguous frontier three; O2 has none. Its subsequent leakage-corrected recurrence study changes partial-start and training-diversity conditions while matching processed-token budget: all four conditions retain frontier three, all have zero three-seed acquisition probability at $K = 4$, and the O0 follow-up gate stays closed. Thus $K_{\text{internal}} = K_{\text{AR}} = 3$ in these phases despite a handcrafted recurrent transition under a different atomic encoding: representability does not imply acquisition.

7 Composition and resource-scheduled closure

For legal set R_K , per- K solvability $\forall K \exists (\mathcal{M}_K, \mathcal{B}_K)$ differs from

$$\exists \mathcal{M} \exists \mathcal{B}(K) \forall K < \infty : \mathcal{M} \text{ solves all } x \in R_K \text{ within } \mathcal{B}(K).$$

Empirically, K_{prefix} is the largest k for which every tested $1 \leq j \leq k$ passes. Report

$$\Sigma = (K_{\text{train}}, K_{\text{test}}, K_{\text{prefix}}, \rho_{\text{step}}, \rho_{\text{stop}}, g, f_{\text{fail}}).$$

Proposition 1 (Finite observations do not establish closure). *For any finite evaluated input–output set consistent with a resource-scheduled procedure, a finite lookup machine agrees on all observations and diverges on an unobserved legal input.*

Proposition 2 (Budgeted local composition). *If D is closed under deterministic $T : D \rightarrow D$, and C exactly preserves state, invokes/integrates T , and applies a total correct stop rule, then $T^k(x)$ is representable whenever all states and trace fit $\mathcal{B}$.*

Independent reliability $r_{\text{stop}} \prod_i r_i$ is only a baseline; acquisition and reliability are separate.

8 Information interfaces and tool algebra

Retrieval changes evidence: $q \mapsto I(q) = D \mapsto F_\theta(q, D)$. Tools are semantic: T_0 lookup, T_1 local transition, T_2 transform, T_3 search/closure, T_4 programmable executor.

	C_0	C_1	C_2	C_3
No tool	one-shot	generation	typed scratchpad	environment loop
T_0	evidence	retrieve/generate	evidence state	iterative retrieval
T_1	one call	serialized calls	current/counter	parent iteration
T_2	delegated transform	calculator/PAL	expression/result	CAS/SQL
T_3	delegated answer	closure call	query/result	search
T_4	run code	generate/execute	program state	code agent

Table 3: Controller $\times$ tool configurations, not an intelligence ladder.

For information I_s , acquisition Q , and execution E , $P(I_s \cap Q \cap E) = P(I_s)P(Q | I_s)P(E | Q, I_s)$. This equals endpoint success only when shortcut branches are excluded. The heterogeneous-network extension is developed separately in Paper 0.2.

9 Classical theory and related work

Topic/work	Pass	Assumptions	Model	Evidence	Boundary here
Hahn; Hao	fixed	attention restrictions	encoder	theory	acquisition
Merrill–Sabharwal; Chiang	fixed	log precision	encoder	theory	learned frontier
Strobl survey	fixed	variant-dependent	both	survey	schema
Merrill–Sabharwal CoT; Li	AR	serial tokens	decoder	theory	acquisition
Joshi; Amiri Bavandpour	AR	iterated/hard attention	decoder	theory	finite learned K
Elhage; Olsson; Meng	fixed activations	trained finite	decoder	mechanistic	causal vs geometric
Dziri; Huang	trained	distribution-specific	decoder	empirical/theory	task controls

Table 4: Topic-organized comparison: assumptions do not transfer across pass, precision, or encoder/decoder setting.

Fixed one-pass. Results depend on attention, precision, positions, and uniformity [1, 2, 3, 4, 5, 6, 7]; universality is representability, not acquisition [8, 9].

Autoregressive/CoT. Serial tokens change expressivity under explicit assumptions [10, 11, 12, 13].

Mechanistic circuits. Residual decomposition, induction heads, and causal tracing distinguish description from necessity [14, 15, 16].

Composition limits. Endpoint success can conceal brittle, distribution-specific computation [17, 18].

10 Late empirical anchors

Paper	Axis	Status/illustration
0.5	internal composition	evidence; non-monotone depth, distributed heads
0.6	semantic composition	v7 gated null; cue-free acquisition fails
0.7	formal inference	P2 passes; F2 fails the three-seed gate
0.8	information/ICL	D5 gated null after valid identification
0.85	AR recurrence	corrected conditions retain $K_{\text{prefix}} = 3$
0.9	tools/harness	pending; no result imported

Table 5: The framework does not depend on these bounded series results.

Paper 0.6 illustrates why a measured frontier must identify the operation being measured. Its v6 `isAncestor` path-length result exposed the candidate directly among path records and therefore measured contextual membership. In v7, two disjoint equal-length chains match candidate exposure and endpoint roles across labels. Across $L \in \{2, 4, 8\}$ and three seeds, mean accuracies are .496, .502, and .498; no model passes even $d = 1$. The cue-free compositional prerequisite fails, all measured frontiers are zero, and $L_{\min}(d)$ is ineligible. This is an acquisition failure in the tested regime, not a representational impossibility.

Paper 0.7 supplies a syntax-sensitive separation. P2, which distinguishes *modus tollens* from contraposition controls, passes at all three seeds (test accuracies .993, 1.000, and .999). F2 unary-functor binding does not: seed 23 reaches only .917 overall and .901 in its worst test template. The registered all-seed, all-template gate therefore stops the deeper unification ladder. Paper 0.8 makes the information/acquisition distinction independently: its D5 generator is exactly identifiable and leakage-free, yet none of 12 learned architecture–seed cells passes any split across 51,456 evaluation rows. Neither negative establishes an architectural lower bound; both block mechanism claims from selectively successful cells.

Finally, Paper 0.85 tests whether serialized output or data intervention moves a learned recurrence boundary. In the corrected recurrence-v2 study, baseline, partial-start, high-diversity, and partial-diversity conditions all have contiguous frontier three over 84 official cells. For every condition, $P_{\text{acquire}}(K = 4) = 0$ across the three seeds. The observed recurrence gain is thus zero in this regime even though the fixed-weight witness above implements reusable transition under its declared encoding. Paper 0.9 remains pending and contributes no empirical premise here.

11 Conjectures and status

	Conjecture	Status
C1	local acquisition need not imply composition	consistent with 0.6 v6/v7 and 0.85
C2	trained models often have bounded one-pass frontiers	consistent with 0.5; 0.6 v7 fails before a frontier
C3	multi-token state expands the measured frontier	not supported through 0.85 recurrence v2; construction is representability only
C4	competent recurrence shifts failure toward control	pending; no learned $K = 4$ recurrence competence
C5	context separates knowledge from procedure	consistent with information-valid 0.8 D5 acquisition null
C6	harnesses may improve reliability before expressivity	pending 0.9

Table 6: Falsifiable conjectures with bounded status language.

12 Discussion and conclusion

Validate identifiability; establish local competence; measure contiguous composition and recurrence frontiers; only then interpret mechanism. The current anchors demonstrate every early stopping point: information can be sufficient while acquisition fails (0.8 D5), a local formal operator can pass while the next syntactic class is seed-unstable (0.7 P2/F2), a shortcut-sensitive frontier can disappear under a cue-free control (0.6 v7), and explicit output recurrence can leave the train-short/test-long boundary unchanged (0.85). Fixed weights calibrate representability, while learned

experiments measure acquisition. Resource-scheduled closure is a quantified property of one reusable procedure, not a long successful trace.

A Fixed-weight micro-Transformer constructions

All implementations use row vectors XW , not WX ; attention is finite $\text{softmax}(QK^\top/\sqrt{d_h})V$. Complete artifacts are under `results/manual_witnesses/`.

A.1 FF-only successor

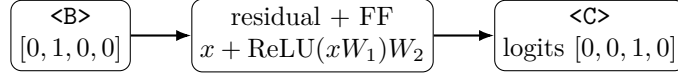


Figure 1: Successor, margin 1.

$$M = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad W_1 = I_4, \quad W_2 = M - I_4 = \begin{bmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

For legal one-hots, $x + \text{ReLU}(xW_1)W_2 = xM$. Result/control: 3/3; SA 0/3; SA+FF 3/3. Biases, positions, and attention are zero.

A.2 SA-only pair lookup

Architecture $L = 1, H = 1, d = 9, d_h = 3$, subspaces [key(3)|query(3)|value/output(3)].

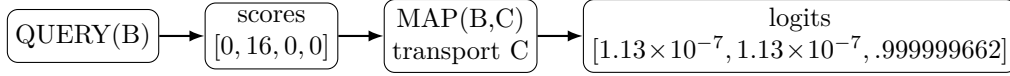


Figure 2: Finite key match and value transport.

Actual sparse blocks: $W_Q[4:6, :] = 27.7128129211I_3 = 16\sqrt{3}I_3$, $W_K[1:3, :] = I_3$, $W_V[7:9, :] = I_3$, $W_O[:, 7:9] = I_3$, others zero. Probabilities are $(1.125351 \times 10^{-7}, .999999662395, 1.125351 \times 10^{-7}, 1.125351 \times 10^{-7})$; margin .999999549859. Result/control: 18/18; FF 6/18 ties; SA+FF 18/18.

A.3 Two-layer SA grandparent

Architecture $L = 2, H = 1, d = 20, d_h = 4$; subspaces are edge key, initial query, parent value, hop state, output.

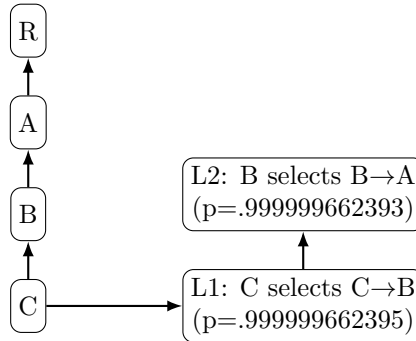


Figure 3: First hop is written to state; layer 2 reads it.

Layer 1 maps query $\rightarrow Q$, edge key $\rightarrow K$, and parent value $\rightarrow V \rightarrow$ hop state; layer 2 maps hop state $\rightarrow Q$, the same edge blocks to K, V , and writes output. Canonical selected probabilities are .999999662395 and .999999662393;

final logits are $(1.125359 \times 10^{-7}, .999999662393, 1.125357 \times 10^{-7}, 0)$, margin .999999549857. Result/control: 2/2; FF and one-layer SA 0/2; SA+FF 2/2.

A.4 Autoregressive root

Architecture $L = 1, H = 1, d = 15, d_h = 5$, reused by an external C_1 model loop.

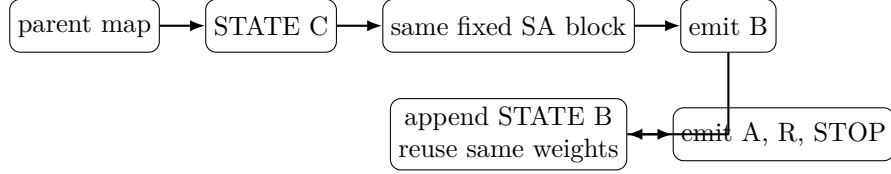


Figure 4: The driver appends argmax tokens but performs no semantic transition.

Step	Current	Target/prediction	Attention (p)	signed margin
1	C	B/B	.999999549860	.999999437324
2	B	A/A	.999999437324	.999999324789
3	A	R/R	.999999324789	.999999212254
4	R	STOP/STOP	.999999212254	.999999099719

Table 7: Tracked canonical trace.

Canonical, reversed, and fixed-shuffle orders each pass all 24 root cases; minimum margin .999999099719. FF passes 6/24 ties; SA+FF 24/24. Cycles, missing/duplicate/conflicting records, non-atomic encodings, larger vocabularies, and out-of-budget lengths are excluded.

A.5 Artifact and formal details

Complete Q/K , raw $QK^\top$, score, probability, V , residual, FF, and logit traces, signed and winner margins, matrices, legal rows, and controls are documented in `manual_transformer_witnesses.md` and generated by `src/cl/manual_transformers/`.

B Probabilistic tasks and better-than-null competence

B.1 Deterministic and probabilistic regimes

Let $P^*(Y | X, D, \mathcal{R})$ be the true conditional law under the declared generator and prior, $q_\theta(Y | X, D)$ the machine distribution, and q_0 an explicit admissible null. A null may be uniform, the class-frequency or majority prior, a query-only predictor $q_0(Y | X)$, a context-shuffled predictor, or a justified task-specific baseline. “Chance” is not a specification. Query-only versus context-aware performance tests whether the machine uses predictive information in D ; mutual information in the generator does not by itself establish such utilization.

Define the null and machine accuracies under their declared decision or sampling rules and

$$A_{\text{Bayes}} = \mathbb{E}_{X,D} \max_y P^*(Y = y | X, D, \mathcal{R}).$$

A useful stochastic predictor may have $A_0 < A_\theta < A_{\text{Bayes}} < 1$. This is better-than-null predictive competence, not deterministic task solution, rule identification, compositionality, recurrence, closure, or a causal mechanism.

Regime	Entropy condition	Primary criterion	Appropriate claim
Identifiable	$H(Y X, D, \mathcal{R}) = 0$	exact accuracy/coverage; ceiling 1	task competence
Probabilistic	$H(Y X, D, \mathcal{R}) > 0$	log loss and accuracy versus null; Bayes ceiling may be < 1	predictive competence
Repeated/ensemble	same underlying task	$A(n, G)$, correlation, cost	compound predictive competence
Oracle best-of- n	either	candidate coverage C_n	coverage only

Table 8: Task/evaluation regimes orthogonal to the six computational axes.

B.2 Proper scoring, calibration, and uncertainty

Accuracy discards distribution quality. For a known generator,

$$\text{CE}(q) = -\mathbb{E} \log q(Y \mid X, D), \quad \Delta_{\text{CE}} = \text{CE}(q_0) - \text{CE}(q_\theta).$$

$\Delta_{\text{CE}} > 0$ means better probabilistic prediction than the null (in nats when log is natural). One may also report $D_{\text{KL}}(P^* \parallel q_\theta)$ only when P^* is known or has a justified estimator. On finite outcomes, NLL, Brier score, and reliability curves complement accuracy; ECE depends on binning and is not required for the deterministic proof/tree experiments.

For illustration, let $P^* = (.50, .25, .15, .10)$, $q_0 = (.25, .25, .25, .25)$, and $q_\theta = (.45, .25, .20, .10)$. The machine assigns outcome 1 probability .45 rather than .25. Independent target and prediction sampling gives match accuracy .3275 versus .25; deterministic argmax accuracy is .50 = A_{Bayes} . In nats, $\text{CE}(q_0) = 1.38629$, $\text{CE}(q_\theta) = 1.21750$, and $\Delta_{\text{CE}} = .16879$. These illustrative numbers expose why accuracy alone misses calibration; they are not experimental results.

Finite evaluations report N and uncertainty, preferably a paired comparison on shared instances; acquisition claims additionally require training seeds. Exhaustive tiny legal domains should use exhaustive accuracy. A capability ladder is

$$\text{null-level} < \text{better-than-null} < \text{Bayes-optimal probabilistic},$$

followed by Bayes-optimal deterministic identification only when the task permits it.

B.3 Repetition, ensembles, and correlated errors

For a declared stochastic mechanism, draw $Y^{(1)}, \dots, Y^{(n)}$ and aggregate

$$\hat{Y}_n = G_n(Y^{(1)}, \dots, Y^{(n)}), \quad A(n, G) = P(\hat{Y}_n = Y).$$

Report both $A(1)$ and $A(n, G)$. Majority/plurality, probability averaging, consensus thresholds, and verifier selection are system-level protocols. Distinguish same-model stochastic repetition, independent trained seeds, architecture/model ensembles, and context/data perturbations; their error correlations differ.

For binary independent runs with odd n and per-run correctness $p > .5$,

$$A_{\text{maj}}(n) = \sum_{j=(n+1)/2}^n \binom{n}{j} p^j (1-p)^{n-j} \longrightarrow 1.$$

This is intuition under iid assumptions, not a forecast for language-model samples; correctness-majority also coincides with label-majority only for binary outcomes or a single wrong label. Runs can share a wrong interpretation, shortcut, training bias, prompt-induced failure, or decoding mode. Useful diagnostics include pairwise answer agreement, correctness correlation, answer-histogram entropy, effective answer count, and the saturation curve $A(n)$. Repeated identical wrong answers are not stronger evidence. If $q_\theta = q_0$, there is no side-information verifier, and aggregation is label-symmetric, repetitions with no complementary information create no missing task information.

B.4 Candidate coverage and deployable selection

Oracle best-of- n measures only

$$C_n = P\left(Y \in \{Y^{(1)}, \dots, Y^{(n)}\}\right),$$

the probability that a correct candidate appears. Deployable accuracy requires an aggregator or verifier to select without target ground truth. High C_n can coexist with poor $A(n, G)$; report both and the selector's cost and information. Repetition plus aggregation, $\mathcal{M}^{(1)}, \dots, \mathcal{M}^{(n)} \rightarrow G$, is already a simple compound computation. Paper 0.2 generalizes this narrow bridge to heterogeneous nodes with different capabilities, costs, reliabilities, and interfaces.

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